

Smart AI Hospital Platform - Complete Development Prompt

Role & Objective

Role: Senior Full-Stack Developer and UI/UX Designer specializing in Medical Tech and Healthcare AI Platforms.

Objective: Create a sophisticated, responsive “Smart AI Hospital” platform that serves as a futuristic healthcare hub where users interact with specialized AI Doctors. The platform should feel like a premium medical technology solution with enterprise-grade design aesthetics inspired by modern healthcare automation platforms.

Tech Stack

Frontend

- **HTML5** with semantic structure
- **Tailwind CSS** (via CDN: <https://cdn.tailwindcss.com>)
- **Vanilla JavaScript** (ES6+) for interactivity
- **Font Awesome** or **Lucide Icons** for iconography
- **Google Fonts:** Inter for body text, Plus Jakarta Sans for headings

Backend

- **Python Flask** for routing and modular page handling
- **Jinja2** templating engine for dynamic content

Design System: “Digital Sanity”

Color Palette

```
/* Primary Colors */

--electric-blue: #0066FF;

--electric-blue-light: #4D94FF;

--electric-blue-dark: #0052CC;

/* Secondary Colors */

--soft-teal: #00C9A7;

--soft-teal-light: #4DD9C0;

--soft-teal-dark: #00A085;

/* Neutral Colors */

--slate-50: #F8FAFC;
```

```
--slate-100: #F1F5F9;

--slate-200: #E2E8F0;

--slate-300: #CBD5E1;

--slate-400: #94A3B8;

--slate-500: #64748B;

--slate-600: #475569;

--slate-700: #334155;

--slate-800: #1E293B;

--slate-900: #0F172A;

/* Accent Colors by Specialty */

--neurology-primary: #6366F1; /* Indigo for Dr. Neuron */

--neurology-dark: #1E1B4B; /* Deep indigo background */

--cardiology-primary: #EF4444; /* Red for Dr. Cardio */

--cardiology-light: #FEE2E2; /* Soft red background */

--vital-primary: #10B981; /* Emerald for Dr. Vital */

--vital-light: #D1FAE5; /* Soft green background */
```

Typography

- **Headings:** Plus Jakarta Sans, weights 600-800
- **Body:** Inter, weights 400-500
- **Hero Title:** 48-64px, font-weight 800, line-height 1.1
- **Section Titles:** 36-42px, font-weight 700
- **Body Text:** 16-18px, line-height 1.6
- **Small/Captions:** 14px, line-height 1.5

Design Patterns

1. **Glassmorphism Cards:** backdrop-blur-xl bg-white/80 border border-white/20 shadow-xl
2. **Rounded Geometry:** Border radius 16px for cards, 12px for buttons, 24px for large containers
3. **Soft Shadows:** shadow-lg shadow-slate-200/50 for elevation
4. **Gradient Accents:** Subtle gradients using primary colors

5. **Micro-interactions:** Hover states, smooth transitions (300ms ease)

Website Structure & Pages

1. Navigation System

Persistent Top Navigation Bar:

- Fixed position, glassmorphism effect on scroll
- Logo: “SmartAI Hospital” with pulse icon animation
- Navigation Links:
 - Dashboard
 - Specialists (dropdown with 3 AI Doctors)
 - Appointments
 - AI Health Vault
- CTA Button: “Start AI Consultation” (electric blue)

Mobile Navigation:

- Hamburger menu with slide-in drawer
- Same links optimized for touch

2. Dashboard Page (Home)

Hero Section:

- Large headline: “Your Personal AI Medical Team, Available 24/7”
- Subheadline: “Connect with specialized AI doctors for instant health insights, symptom checking, and personalized care recommendations.”
- Two CTAs: “Meet Our Specialists” (primary) | “How It Works” (secondary/outline)
- Background: Subtle animated gradient mesh (soft-teal to electric-blue)
- Floating elements: Abstract medical icons with gentle float animation

Quick AI Triage Search Bar:

- Prominent search input with placeholder: “Describe your symptoms or health concern...”
- AI sparkle icon on left
- “Analyze with AI” button on right
- Glassmorphism container with glow effect on focus

Active Health Monitoring Stats (4-Column Grid):

[Icon]	[Icon]	[Icon]	[Icon]
50,000+	99.8%	< 2min	24/7
Consultations Completed	Accuracy Rate	Response Time	Available

Meet Your AI Specialists Section:

- Section title: “Meet Your AI Medical Specialists”
- 3 specialist cards in a row (responsive: 1 col mobile, 3 col desktop)
- Each card contains:
 - AI Avatar placeholder (circular, with specialty color ring)
 - Doctor name & title
 - Specialty badge
 - Short description (2 lines)
 - “Consult Now” button
 - Hover effect: Card lifts, avatar scales slightly

How It Works Section:

- 4-step horizontal process with connecting lines
- Step 1: Describe Symptoms → Step 2: AI Analysis → Step 3: Specialist Match → Step 4: Get Insights
- Each step has icon, title, and brief description

Trust & Security Section:

- Compliance badges: HIPAA, SOC 2, GDPR
- Security statement: “Your health data is encrypted and protected”
- Testimonial carousel with patient reviews

Footer:

- 4-column layout: Brand, Platform, Specialists, Contact
- Social media links
- Legal links: Privacy Policy, Terms of Service, HIPAA Notice

3. Specialist Directory Page

Page Layout:

- Page header: “AI Medical Specialists”
- Subtitle: “Choose a specialist based on your health concerns”
- Filter tabs: All | Neurology | Cardiology | General Wellness

Specialist Cards (Detailed):

Avatar Dr. Neuron
AI Neurology Specialist

[Neurology Badge]

Advanced neural network analysis
for brain health, migraines, and
neurological concerns.

[View Profile] [Start Chat]

4. Individual Specialist Pages (Dynamic)

Route Structure: /specialist/<doctor_id>

Dr. Neuron - Neurology Specialist (Dark Theme):

- Background: Deep indigo gradient with subtle neural network pattern animation
- Color scheme: Indigo (#6366F1) accents on dark background
- Avatar: AI doctor with neural network overlay effect
- Specialty icon: Brain/neural network
- Greeting: "Hello, I'm Dr. Neuron. I'm here to help with any neurological concerns."

Dr. Cardio - Heart Health Specialist:

- Background: Soft red gradient with subtle pulse wave animation at bottom
- Color scheme: Red (#EF4444) accents
- Avatar: AI doctor with heartbeat EKG overlay
- Specialty icon: Heart with pulse
- Greeting: "Hello, I'm Dr. Cardio. Let's take care of your heart health together."

Dr. Vital - General Wellness Specialist:

- Background: Soft green gradient with organic flowing shapes
- Color scheme: Emerald (#10B981) accents
- Avatar: AI doctor with wellness/leaf elements
- Specialty icon: Heart with plus sign
- Greeting: "Hello, I'm Dr. Vital. I'm here to support your overall wellness journey."

Common Specialist Page Structure:

[Back to Specialists]

Dr. [Name]
AI Avatar [Specialty] Specialist
(Large) [Status: Online]

[Start Consultation] [View Credentials]

TABS: [About] [Chat] [Diagnostic Capabilities]

TAB: About

- Detailed description of the AI specialist
- Areas of expertise (bullet list)
- Training data & certifications

TAB: Chat Interface (UI Only)

[AI Avatar] Hello! How can I assist you today?

[User message placeholder]

[Send] [Attach]

Type your health concern...

TAB: Diagnostic Capabilities

Grid layout of capability cards:

[Icon]	[Icon]	[Icon]	[Icon]
Symptom	Risk	Treatment	Health
Analysis	Assessment	Suggestions	Monitoring

5. Appointments Page

Layout:

- Calendar view (monthly/weekly toggle)
- Upcoming appointments list
- “Book New Appointment” button
- Appointment cards with:
 - Specialist avatar
 - Date & time
 - Status (confirmed/pending/completed)
 - Join/Reschedule/Cancel actions

6. AI Health Vault Page

Layout:

- Health summary dashboard
- Recent consultations history
- Saved health insights
- Health metrics tracking (if applicable)
- Download health reports functionality

Flask Application Structure

/smart-ai-hospital/

```
app.py # Main Flask application
requirements.txt # Python dependencies
static/
  css/
    custom.css # Custom styles beyond Tailwind
  js/
    main.js # Main JavaScript functionality
    typing-effect.js # AI greeting typing animation
    chat-interface.js # Chat UI interactions
  images/
    logo.svg # SmartAI Hospital logo
    avatars/ # AI Doctor avatars
      dr-neuron.png
      dr-cardio.png
      dr-vital.png
    icons/ # Custom icons

templates/
  layout.html # Base template with common structure
  index.html # Dashboard/Home page
  specialists.html # Specialist directory
  specialist.html # Dynamic specialist page template
  appointments.html # Appointments page
  health-vault.html # AI Health Vault page
```

app.py Routes

Required routes:

```
@app.route('/') # Dashboard/Home
@app.route('/specialists') # Specialist directory
@app.route('/specialist/<name>') # Individual specialist (dynamic)
@app.route('/appointments') # Appointments page
@app.route('/health-vault') # AI Health Vault
```

Specialist Data Structure

```
SPECIALISTS = {
  'dr-neuron': {
    'id': 'dr-neuron',
    'name': 'Dr. Neuron',
    'title': 'AI Neurology Specialist',
    'specialty': 'Neurology',
    'theme': 'dark',
    'primary_color': '#6366F1',
    'description': 'Advanced neural network analysis for brain health...',
    'greeting': "Hello, I'm Dr. Neuron. I'm here to help with any neurological concerns.",
    'expertise': [
      'Headache and migraine analysis',
      'Cognitive function assessment',
      'Sleep disorder evaluation',
      'Neuropathy symptom checking'
    ],
    'capabilities': [
      {'icon': 'brain', 'title': 'Symptom Analysis', 'desc': 'Analyze neurological symptoms'},
      {'icon': 'activity', 'title': 'Risk Assessment', 'desc': 'Evaluate potential risks'},
      {'icon': 'clipboard', 'title': 'Treatment Info', 'desc': 'Provide treatment guidance'},
      {'icon': 'trending-up', 'title': 'Progress Tracking', 'desc': 'Monitor symptom changes'}
    ]
  },
  'dr-cardio': {
    'id': 'dr-cardio',
    'name': 'Dr. Cardio',
    'title': 'AI Cardiology Specialist',
    'specialty': 'Cardiology',
    'theme': 'light',
    'primary_color': '#EF4444',
    'description': 'Comprehensive heart health analysis and cardiovascular care...',
    'greeting': "Hello, I'm Dr. Cardio. Let's take care of your heart health together.",
    'expertise': [
      'Heart rhythm analysis',
      'Blood pressure guidance',
      'Chest pain evaluation',
      'Cardiovascular risk assessment'
    ],
    'capabilities': [
      {'icon': 'heart', 'title': 'Heart Analysis', 'desc': 'Evaluate cardiac symptoms'},
      {'icon': 'activity', 'title': 'ECG Insights', 'desc': 'Interpret heart rhythm data'},
      {'icon': 'shield', 'title': 'Risk Screening', 'desc': 'Assess cardiovascular risks'},
      {'icon': 'droplet', 'title': 'BP Guidance', 'desc': 'Blood pressure management'}
    ]
  },
  'dr-vital': {
```



```

    'id': 'dr-vital',
    'name': 'Dr. Vital',
    'title': 'AI General Wellness Specialist',
    'specialty': 'General Wellness',
    'theme': 'light',
    'primary_color': '#10B981',
    'description': 'Holistic health analysis for overall wellness and preventive care...',
    'greeting': "Hello, I'm Dr. Vital. I'm here to support your overall wellness journey.",
    'expertise': [
      'General health checkups',
      'Preventive care guidance',
      'Nutrition and lifestyle advice',
      'Mental wellness support'
    ],
    'capabilities': [
      {'icon': 'user-check', 'title': 'Health Check', 'desc': 'Comprehensive wellness review'},
      {'icon': 'apple', 'title': 'Nutrition Guide', 'desc': 'Dietary recommendations'},
      {'icon': 'sun', 'title': 'Lifestyle Tips', 'desc': 'Healthy living advice'},
      {'icon': 'smile', 'title': 'Mental Wellness', 'desc': 'Stress and anxiety support'}
    ]
  }
}

```

Key Features to Implement

1. Navigation & Routing

- Smooth page transitions using JavaScript
- Active state highlighting in navigation
- Breadcrumb navigation for specialist pages

2. AI Typing Effect

```

// Typewriter effect for AI greetings
function typeWriter(element, text, speed = 50) {
  let i = 0;
  element.innerHTML = '';
  function type() {
    if (i < text.length) {
      element.innerHTML += text.charAt(i);
      i++;
      setTimeout(type, speed);
    }
  }
  type();
}

```

3. Chat Interface (UI Only)

- Message bubbles (user right, AI left)
- Typing indicator animation
- Quick reply suggestion chips
- File attachment button (visual only)
- Auto-scroll to latest message

4. Responsive Design

- Mobile-first approach
- Breakpoints: sm (640px), md (768px), lg (1024px), xl (1280px)
- Touch-friendly buttons (min 44px)
- Collapsible navigation on mobile

5. Animations & Interactions

- Scroll-triggered fade-in animations for sections
- Hover effects on cards (lift + shadow)
- Pulse animation on online status indicators
- Smooth tab switching
- Loading skeleton screens

6. Glassmorphism Effects

```
.glass-card {  
  background: rgba(255, 255, 255, 0.8);  
  backdrop-filter: blur(20px);  
  
  -webkit-backdrop-filter: blur(20px);  
  
  border: 1px solid rgba(255, 255, 255, 0.3);  
  border-radius: 16px;  
  box-shadow: 0 8px 32px rgba(0, 0, 0, 0.1);  
}
```

Tailwind Configuration

Include this in your HTML head:

```
<script src="https://cdn.tailwindcss.com"></script>  
<script>  
  tailwind.config = {  
    theme: {  
      extend: {
```

```

        colors: {
            'electric-blue': '#0066FF',
            'electric-blue-light': '#4D94FF',
            'soft-teal': '#00C9A7',
            'neurology': '#6366F1',
            'cardiology': '#EF4444',
            'vital': '#10B981',
        },
        fontFamily: {
            'heading': ['Plus Jakarta Sans', 'sans-serif'],
            'body': ['Inter', 'sans-serif'],
        },
        animation: {
            'float': 'float 6s ease-in-out infinite',
            'pulse-slow': 'pulse 3s ease-in-out infinite',
            'typing': 'typing 2s steps(40, end)',
        },
        keyframes: {
            float: {
                '0%, 100%': { transform: 'translateY(0)' },
                '50%': { transform: 'translateY(-20px)' },
            }
        }
    }
}
}
}
</script>

```

Required HTML Template Structure

layout.html (Base Template)

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>{% block title %}Smart AI Hospital{% endblock %}</title>

    <!-- Fonts -->
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600&family=Plus+
        Jakarta+Sans:wght@600;700;800&display=swap" rel="stylesheet">

    <!-- Tailwind CSS -->
    <script src="https://cdn.tailwindcss.com"></script>
    <script>
        tailwind.config = {
            theme: {
                extend: {

```

```

        colors: {
            'electric-blue': '#0066FF',
            'soft-teal': '#00C9A7',
            'neurology': '#6366F1',
            'cardiology': '#EF4444',
            'vital': '#10B981',
        },
        fontFamily: {
            'heading': ['Plus Jakarta Sans', 'sans-serif'],
            'body': ['Inter', 'sans-serif'],
        }
    }
}

</script>

<!-- Custom CSS -->
<link rel="stylesheet" href="{url_for('static', filename='css/custom.css')}">

{% block extra_css %}{% endblock %}
</head>
<body class="font-body bg-slate-50 text-slate-800">
    <!-- Navigation -->
    {% include 'partials/navbar.html' %}

    <!-- Main Content -->
    <main>
        {% block content %}{% endblock %}
    </main>

    <!-- Footer -->
    {% include 'partials/footer.html' %}

    <!-- JavaScript -->
    <script src="{url_for('static', filename='js/main.js')}"></script>
    {% block extra_js %}{% endblock %}
</body>
</html>

```

Deliverables Checklist

Files to Create:

1. **app.py** - Flask application with all routes
2. **templates/layout.html** - Base template
3. **templates/partials/navbar.html** - Navigation component
4. **templates/partials/footer.html** - Footer component
5. **templates/index.html** - Dashboard/Home page
6. **templates/specialists.html** - Specialist directory
7. **templates/specialist.html** - Dynamic specialist page
8. **templates/appointments.html** - Appointments page

9. **templates/health-vault.html** - AI Health Vault page
10. **static/css/custom.css** - Custom styles
11. **static/js/main.js** - Main JavaScript
12. **static/js/typing-effect.js** - Typing animation
13. **static/js/chat-interface.js** - Chat UI functionality
14. **requirements.txt** - Python dependencies

Features to Verify:

- ☐ Responsive navigation with glassmorphism effect
- ☐ Hero section with animated gradient background
- ☐ Quick AI Triage search bar with focus glow
- ☐ Stats section with animated counters
- ☐ 3 specialist cards with hover effects
- ☐ How It Works section with step connectors
- ☐ Dynamic specialist pages with unique themes
- ☐ Chat interface with typing effect
- ☐ Tab navigation on specialist pages
- ☐ Diagnostic capabilities grid
- ☐ Smooth scroll animations
- ☐ Mobile-responsive design
- ☐ Footer with all required links

Quality Standards

1. **Code Quality:** Clean, well-commented, PEP 8 compliant Python
2. **Accessibility:** ARIA labels, keyboard navigation, color contrast WCAG 2.1 AA
3. **Performance:** Optimized images, minimal external dependencies
4. **SEO:** Proper meta tags, semantic HTML, structured data
5. **Cross-browser:** Works on Chrome, Firefox, Safari, Edge (latest 2 versions)

Final Instructions

Create a complete, production-ready “Smart AI Hospital” platform that:

1. Follows the “Digital Sanity” design aesthetic
2. Implements all specified pages and features
3. Uses Flask for backend with proper routing
4. Features unique, themed specialist pages
5. Includes smooth animations and interactions
6. Is fully responsive across all devices
7. Has a professional, trustworthy healthcare appearance

The platform should feel like a premium medical technology solution that patients would trust with their health information.